

Solution Requirements

SMARTBRIDGE - Solution Requirements

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-6970
- **Team Size:** 3
- **Team Leader:** Prince Kumar P
- **Team Members:** Gowtham L, Guru Moorthi M
- **Project Name:** AI FitTrack
- **Maximum Marks:** 4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register using email and password. Passwords are encrypted using <code>bcryptjs</code> . Login issues a signed JWT token.	High
2	Authorization levels	Authenticated Users can manage their personal workouts and invoke AI features. Middleware restricts unauthorized API calls.	High
3	External interfaces	Integrates with Google Gemini AI API using the <code>@google/genai</code> SDK for workout generation and analytics.	High
4	Transactions processing	Handles transactional CRUD requests for workout records (create, read, search, update, delete) returning JSON responses.	High
5	Reporting	Generates performance reports including total workouts, average	Medium

		duration, and total calories burned via AI insights.	
6	Business rules, etc.	Validation rules: Workout duration must be > 0, calories burned $\geq$ 0, and emails must be unique.	High
7	Compliance to laws	Ensures user privacy by salting/hashing passwords and avoiding exposure of credentials in API payloads.	High
8	Other	Full-text query searching for workouts based on workout name, category, or workout date.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category PDF	Requirement Description PDF	Target Metric / Acceptance Criteria PDF
1	Performance & Speed	Fast response times for CRUD APIs and Gemini AI request processing.	API responses < 200ms ; AI generation < 3seconds.
2	Scalability	Handles growing database records and concurrent user requests.	MongoDB Atlas cluster scales seamlessly up to 10,000 users.
3	Security & Data Privacy	Secure transmission, encrypted stored passwords, and protected API endpoints.	Passwords hashed with bcrypt (salt 10); JWT authentication required.

4	Reliability & Availability	High server uptime and resilient database connectivity.	\$99.9\%\$ uptime per month with automatic MongoDB Atlas connection retry.
5	Usability & Accessibility	Standard RESTful API output structure and clear error responses.	\$100\%\$ standardized JSON format with status codes (200, 201, 400, 500).
6	Maintainability	Clean Code separation following the MVC architecture.	Modular separation across Controllers, Services, Models, and Routes.